

IFC Power Control via RIBBO

Appendix / Backup Slides — June 2026

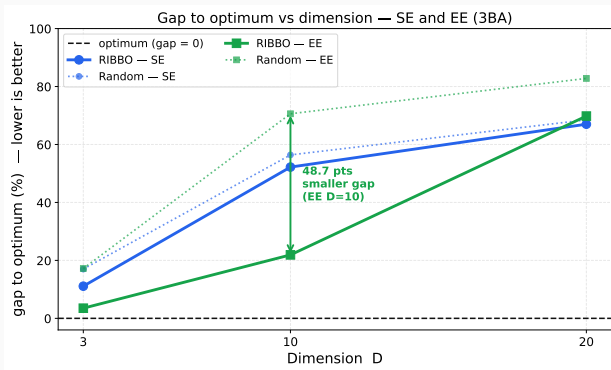
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The whole story in one view (3BA recipe, gap to optimum)

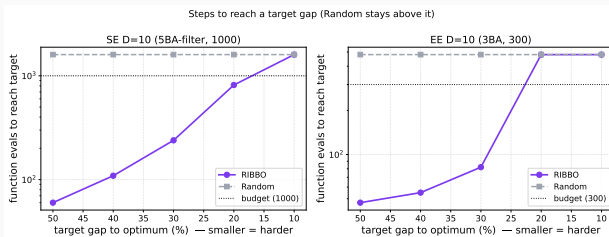
- EE $D=3$: smallest gap (3.5%); gap grows with D for both
- **RIBBO's gap far below Random at EE $D=10$** (48.7 pts smaller)
- SE: RIBBO and Random gaps stay close (on/off solutions are simple)
- $D=20$: large gap ($\sim 70\%$) — recipe/data limit (budget doesn't fix it)

“why RIBBO” = green solid-vs-dotted gap (EE, mid/high D).



How fast does each method shrink the gap to optimum?

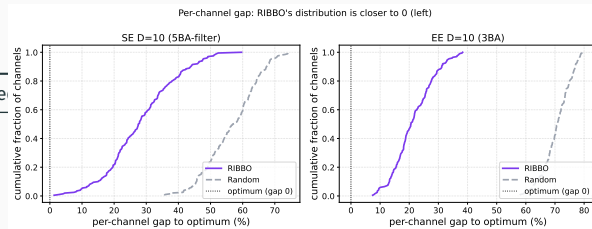
- RIBBO inference is cheap $\Rightarrow$ “just run longer”
- **SE $D=10$:** RIBBO gets the gap below 20% by ~ 815 evals; Random stays above 50% within the budget ($\sim 53\%$)
- **EE $D=10$:** RIBBO gets the gap below 30% by ~ 82 evals; Random stays above 50% ($\sim 71\%$)
- **SE $D=3$:** RIBBO $9\times$ faster to a 30% gap; reaches a 10% gap where Random stalls at 13%



Random pinned above the budget line = does not reach that target within the budget.

Win-rate & distribution over channels (300-step horizon)

Case	win-rate	R med gap	Rnd me
SE $D=3$	73.6%	3.7	
SE $D=10$ (5BA)	100%	27.7	
EE $D=3$	88.5%	2.0	
EE $D=10$	100%	20.6	
SE $D=20$ (4BA)	99.5%	58.5	

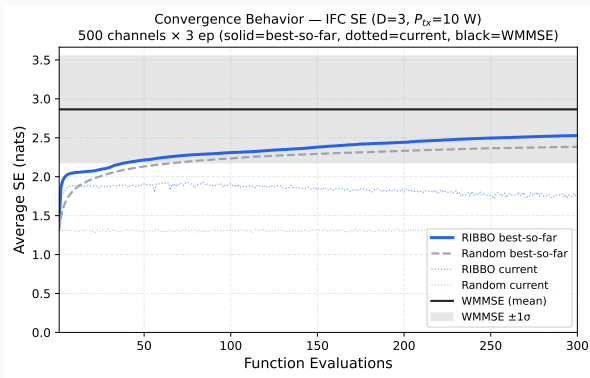


CDF shifted left = more channels with small gap (closer to optimum).

- win-rate = % of channels where RIBBO's gap is smaller than Random's
- At $D=10$ RIBBO has a smaller gap than Random on **every** channel (SE and EE)

Our WMMSE is the true optimum

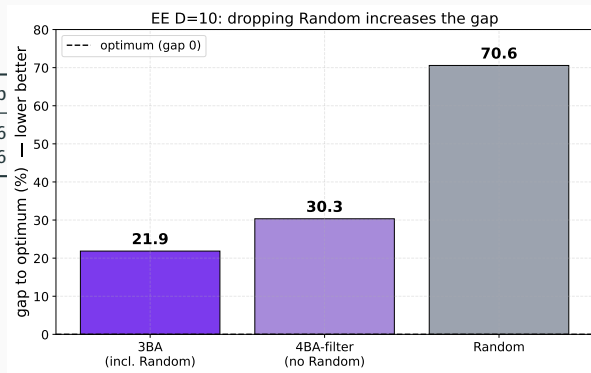
- Shi 2011 box-constrained WMMSE, 10 restarts
- Matches brute-force grid: gap 0/10 channels
- Our value 2.864 nats vs Lee 2.50 (channel-distribution / setup difference; we use the *same* channels for a fair RIBBO-vs-WMMSE comparison)
- RIBBO 3BA gets within **11.1% gap** of WMMSE



For EE, dropping Random *lowers performance*

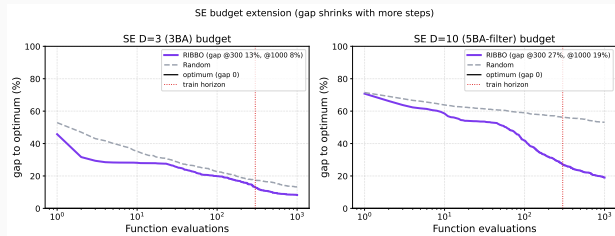
EE $D=10$ model	RIBBO gap	Random gap
3BA (incl. Random)	21.9	70.6
4BA-filter (no Random)	30.3	70.6

- Opposite of SE high- D (where curation *shrinks* the gap 23.6 pts)
- EE optimum is interior/low-power; Random's broad low-power coverage is genuinely useful
- $\Rightarrow$ don't reuse the SE recipe blindly for EE



Run longer $\Rightarrow$ smaller gap (inference is cheap)

- SE $D=3$ 3BA: gap 13.0 $\rightarrow$ **8.3** (−4.8 pts) at 1000 steps
- SE $D=10$ 5BA-filter: gap 27.1 $\rightarrow$ **19.0** (−8.1 pts)
- Higher dimension $\Rightarrow$ larger budget gain



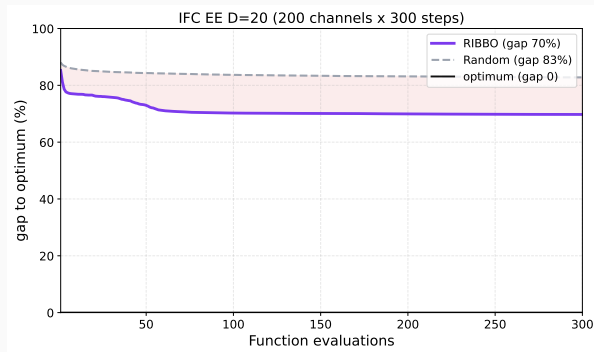
Objective	Setting	RIBBO gap	Random gap	RIBBO saves
SE	$D=3$ (3BA)	11.1	17.0	5.9
SE	$D=10$ (3BA)	52.2	56.4	4.2
SE	$D=10$ (5BA-filter)	28.6	56.4	27.8
SE	$D=20$ (3BA)	67.0	68.5	1.5
SE	$D=20$ (4BA-filter)	58.1	68.5	10.4
EE	$D=3$ (3BA)	3.5	17.2	13.7
EE	$D=10$ (3BA)	21.9	70.6	48.7
EE	$D=10$ (4BA-filter)	30.3	70.6	40.3

gap to optimum (%), lower better. Budget (1000 steps): SE $D=3$ gap $\rightarrow$ 8.3, SE $D=10$ gap $\rightarrow$ 19.0. Win-rate $D=10$: SE & EE both 100% of channels.

The EE advantage persists at $D=20$

EE $D=20$ (3BA)	gap to opt	95% CI
RIBBO	69.8	± 0.8
Random	82.8	± 0.3

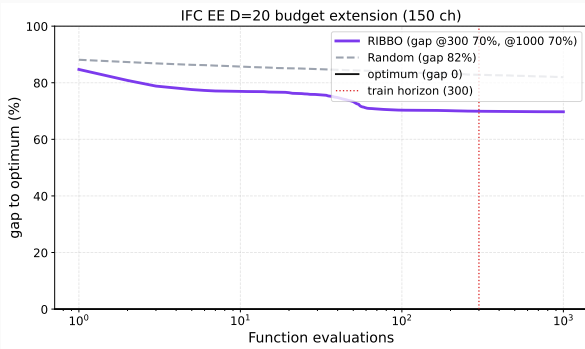
- RIBBO's gap is **13.0 pts smaller, 100% win-rate**
- Bigger lead than SE $D=20$ (13.0 vs 10.4 pts smaller) — the interior EE optimum keeps RIBBO valuable
- Gap grows with D (fixed 3BA recipe). **Budget does not shrink it** (next slide) — a data/recipe limit, not budget



Running longer (300 $\rightarrow$ 1000 steps) gives almost no gain here

- RIBBO gap @300 = 69.9% $\rightarrow$ @1000 = 69.7% (−0.2 pts)
- Contrast: SE budget shrank the gap 8.1 pts ($D=10$), 4.8 ($D=3$)
- RIBBO's gap plateaus high $\Rightarrow$ the model has saturated, not under-budgeted

Refines the “recovery levers” claim:
budget helps only while the model is still improving (SE). EE $D=20$'s gap is a **data-capacity limit** — it needs more channels/data, not more steps.

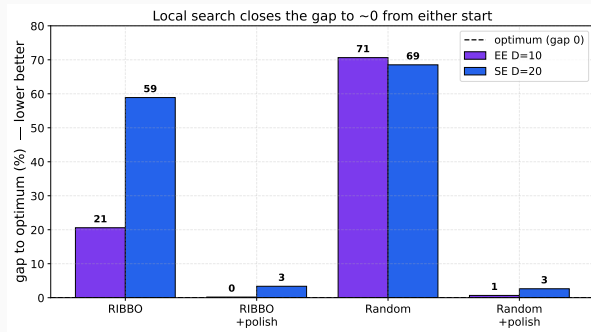


Flat past ~ 50 evals; train-horizon line at 300 makes no difference.

A cheap local search reaches $\sim$ optimum from *either* starting point

gap to optimum	EE $D=10$	SE $D=20$
RIBBO	20.6	58.9
RIBBO + polish	0.2	3.4
Random	70.6	68.5
Random + polish	0.7	2.6

- Polish dominates; the init barely matters (RIBBO $\approx$ Random after polish)
- $\Rightarrow$ with the channel model/gradients a local search reaches the optimum (as WMMSE/FP do); RIBBO's value is the **no-model** case



Obj.	Setting	RIBBO gap ($\pm$ CI)	Random gap ($\pm$ CI)	win-rate
SE	$D=3$ (3BA)	10.7 ± 1.3	17.8 ± 1.0	73.6%
SE	$D=10$ (5BA-filter)	28.4 ± 1.5	56.7 ± 1.2	100%
SE	$D=20$ (4BA-filter)	58.0 ± 1.0	68.5 ± 0.7	99.5%
EE	$D=3$ (3BA)	3.2 ± 0.5	17.6 ± 1.7	88.5%
EE	$D=10$ (3BA)	21.5 ± 1.0	70.6 ± 0.7	100%
EE	$D=20$ (3BA)	69.8 ± 0.8	82.8 ± 0.3	100%

gap to optimum (%), lower better. $\pm$ is 95% CI ($1.96 \text{ sd} / \sqrt{n}$), $n=200$. 500-channel re-eval confirms: EE $D=10$ 20.7 ± 0.7 , SE $D=20$ 58.6 ± 0.6 . Intervals are narrow $\Rightarrow$ not sampling artifacts.